

D-dimer Testing for Suspected Pulmonary Embolism in Outpatients

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The plasma level of D-dimer, a fibrin degradation product (FDP), is nearly always increased in the presence of acute pulmonary embolism (PE). Hence, a normal D-dimer level (below a cutoff value of 500 $\mu\text{g/L}$ by enzyme-linked immunosorbent assay [ELISA]) may allow the exclusion of PE. To assess the negative predictive value of a D-dimer concentration below 500 $\mu\text{g/L}$ in outpatients with suspected PE, and the safety of withholding anticoagulant treatment from such patients, we performed D-dimer assays, lower limb venous compression ultrasonography, and lung scans in 671 consecutive outpatients presenting in the Emergency Center of the Geneva University Hospital with suspected PE. Pulmonary angiography was reserved for patients with an inconclusive noninvasive workup. Patients with a normal D-dimer concentration were discharged without anticoagulant treatment and followed for 3 mo. The prevalence of PE was 29%, and D-dimer (using a cutoff of 500 $\mu\text{g/L}$) had a diagnostic sensitivity for PE of 99.5%. Overall diagnostic specificity of D-dimer was 41%, but it was lower among older patients. Of the 198 patients with a D-dimer concentration below the cutoff value, 196 were free of PE, one had a PE, and one had incomplete information because of loss to follow-up. Thus, the negative predictive value of D-dimer concentration fell between 197 of 198 and 196 of 198 cases of PE (99% [95% CI: 96.4 to 99.9]). Using a cutoff value of 4,000 $\mu\text{g/L}$, the overall specificity of D-dimer concentration for PE was 93.1%. In conclusion, a plasma D-dimer concentration below 500 $\mu\text{g/L}$ allows the exclusion of PE in 29% of outpatients suspected of having PE. Withholding anticoagulation from such patients is associated with a conservative 1% risk of thromboembolic events during follow-up. Perrier A, Desmarais S, Goehring C, de Moerloose P, Morabia A, Unger P-F, Slosman D, Junod A, Bounameaux H. D-dimer testing for suspected pulmonary embolism in outpatients.

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The diagnosis of pulmonary embolism (PE) remains a diagnostic challenge. Pulmonary angiography, admittedly the "gold standard" technique for this diagnosis (1), is costly, invasive (2), and not universally available (3). Noninvasive procedures such as lung scan (4-6) and lower limb venous compression ultrasonography (7, 8) have certainly simplified the diagnostic approach. However, lung scan is diagnostic in only 30 to 50% of patients (4, 6, 8), and many institutions lack nuclear medicine facilities. Ultrasonography reveals deep-vein thrombosis in 60% of patients with PE, but only in 8 to 19% of patients with a nondiagnostic lung scan (8-12).

Plasma D-dimer, a degradation product of crosslinked fibrin, has been extensively studied in the setting of suspected venous thromboembolism (deep-vein thrombosis and PE)

(13-15). At the usual cutoff value (500 $\mu\text{g/L}$), the specificity of D-dimer for PE is low (20 to 50%) owing to the number of conditions that activate the coagulation and fibrinolytic processes, making it ill-suited for establishing a diagnosis of PE (14, 15). On the other hand, when measured with an enzyme-linked immunosorbent assay (ELISA), the plasma level of D-dimer is nearly always increased in acute PE (sensitivity: 97%), with a normal D-dimer level virtually ruling out the disease (14, 15). Despite the potential for D-dimer as a screening analyte for allowing the exclusion of PE in a significant proportion of patients, this test has not yet gained wide acceptance in this setting. Published studies have been criticized for lack of standardization of D-dimer assays, small numbers of patients, the lack of a comprehensive diagnostic standard, and insufficient information on patient selection (15, 16). Most important has been the call for a prospective assessment of clinical decisions based on D-dimer test results (15, 16).

In a previous trial of the safety and efficacy of a sequential noninvasive diagnostic strategy for PE in 308 outpatients (8), we showed the high sensitivity (99%) of an ELISA D-dimer assay. Moreover, among the 199 patients without PE, of whom 100 had a normal D-dimer concentration, the risk of thromboembolic events during the 3-mo follow-up period was only 1.0% (95% CI: 0.1 to 3.6). To increase the statistical power of

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our analysis, we extended this observation to 363 additional consecutive outpatients suspected of having PE, who were managed according to their D-dimer result, resulting in a cohort of 671 outpatients. In both series, patients with a normal D-dimer (a concentration below 500 $\mu\text{g/L}$ by ELISA) were not anticoagulated. Because untreated PE will result in a high frequency of recurrent PE (17, 18), all patients were followed for 3 mo to assess the risk of venous thromboembolic events.

METHODS

Patients

Seven hundred forty-two outpatients presenting with clinically suspected PE at the emergency center of the University Hospital of Geneva between October 1992 and March 1995, and undergoing lung scan, were included in a prospective study of PE diagnosis, of which an earlier report has been published (8). The University Hospital of Geneva is both a teaching and a general hospital. The patients enrolled in the study were seen directly in the emergency ward or were referred by a primary-care physician. PE was suspected either by the primary care or the emergency ward physician. The clinical probability of PE was assessed by the emergency ward physician, one of approximately 70 internal medicine residents in training during the study.

Ineligibility and exclusion criteria were identical in this and in the previous series (8). Ineligibility criteria were age younger than 16 yr; referral from another institution for workup of suspected PE occurring during a hospital stay ($n = 6$); symptoms of deep-vein thrombosis as the principal presenting complaint ($n = 2$); occurrence of deep-vein thrombosis less than 3 mo before inclusion in the protocol ($n = 5$); refusal of consent ($n = 2$); impossibility of follow-up ($n = 4$); or lung scan refused or not done ($n = 3$). Twenty-two patients (3.0%) were ineligible.

Patients whose lung scan was read in comparison with a previous scan ($n = 21$), a situation in which the Prospective Investigation of Pulmonary Embolic Disease (PIOPED) criteria (4) may not apply, were excluded from the study. Since both this and the previous study (8) included clinical probability in the diagnostic strategy, patients who were not assigned a figure for clinical probability of PE ($n = 25$) were also excluded. D-dimer was not measured in three patients. Therefore, the study population available for analysis consisted of 671 patients.

The protocol for the study was approved by the Ethics Committee of the Department of Medicine of the University of Geneva, and written informed consent was obtained from all patients.

Study Design

The study was designed as a prospective outcome trial with a 3-mo follow-up. Patients underwent clinical evaluation in the emergency center prior to undergoing any other test, and clinical probability of PE was rated between 0 and 100%, as detailed elsewhere (8). Lung scan, D-dimer assay, and lower-limb venous-compression ultrasonography were done on all patients, pulmonary angiography being done only in patients with an inconclusive noninvasive workup. Noninvasive workup was completed within 24 h.

Patients were considered not to have PE when the following criteria were met: negative D-dimer assay result (D-dimer level below 500 $\mu\text{g/L}$); normal lung scan; or normal pulmonary angiogram. Because suspected PE is managed in our institution according to a decision-analysis-based strategy (8) involving an assessment of clinical probability of disease, patients with a low clinical probability (as defined previously [8]) and a nondiagnostic lung scan were also considered not to have PE. Patients without PE according to the diagnostic workup were not anticoagulated, and the diagnostic standard against which this diagnosis was compared was the risk of thromboembolic events over the 3-mo follow-up period (8, 19). Diagnostic criteria for PE were a high-probability lung scan; deep-vein thrombosis shown by ultrasonography and a nondiagnostic lung scan; or a pulmonary angiogram showing an embolus. Only a few patients ($n = 15$) were diagnosed as having PE on the basis of a high clinical probability and a nondiagnostic

lung scan. These patients were judged too ill to be subjected to pulmonary angiography.

Three-month Follow-up

All adverse events occurring during the 3-mo follow-up period were recorded: venous thromboembolic events (deep-vein thrombosis or PE), deaths, and severe hemorrhages. Diagnoses of venous thromboembolic events were established with usual criteria (deep-vein thrombosis: abnormal findings on compression ultrasonography or venography; PE; high-probability ventilation-perfusion scan or abnormal angiogram). The follow-up procedure has been described elsewhere (8).

Diagnostic Studies

The techniques for performing lung scan and pulmonary angiography and assaying plasma D-dimer (Asserachrom D-Di enzyme immunoassay kit; Diagnostica Stago, Asnières-sur-Seine, France) have been described elsewhere (13). Lung scans were classified as normal or of very low, low, intermediate, or high probability according to the PIOPED interpretation criteria (4). Because the D-dimer assay is provided as a multitest batch kit, the assay was performed once a day by a technician in our hemostasis laboratory who was unaware of the clinical probability of PE, the lung scan result, or the final diagnosis. Each patient had only one D-dimer assay before undergoing lung scan.

Lower limb B-mode venous compression ultrasonography was performed within 24 h of patient entry into the study in all cases, by trained staff (four persons throughout the study) in a blinded fashion. It consisted of a real-time B-mode examination of the common femoral and popliteal veins. The criterion for diagnosing deep-vein thrombosis was noncompressibility of the vein (7).

Statistical Analysis

Sensitivity and specificity of D-dimer, ultrasonography, and pulmonary angiography for PE were calculated in reference to the final diagnosis (20). The exact 95% CIs for proportions were calculated from the binomial distribution with the Confidence Interval Analysis (CIA) computer software (21).

RESULTS

The series included 671 outpatients. The median age was 59 yr (range: 17 to 92 yr) for the group without PE ($n = 475$) and 70 yr (range: 19 to 95 yr) for the group with PE ($n = 196$). The prevalence of PE in this population (29%), as well as the proportion of normal and high-probability lung scans, is consis-

TABLE 1
RESULTS OF DIAGNOSTIC STRATEGY

	Patients (n)
Pulmonary embolism	
High probability lung-scan alone	47
Deep-vein thrombosis* and high-probability lung scan	66
Deep-vein thrombosis* and nondiagnostic [†] lung scan	34
Abnormal pulmonary angiogram	34
High clinical probability and nondiagnostic lung scan	15
Total	196
No pulmonary embolism	
Normal lung scan alone	39
Normal lung scan and D-dimer < 500 $\mu\text{g/L}$	69
D-dimer < 500 $\mu\text{g/L}$ and nondiagnostic lung scan	127
Normal pulmonary angiogram	130
Low clinical probability and very-low- or low-probability lung scan	110
Total	475

* Deep-vein thrombosis diagnosed through lower-limb venous-compression ultrasonography.

[†] Nondiagnostic scan (i.e., abnormal but non-high-probability scan).

TABLE 2
RESULTS OF D-DIMER MEASUREMENT ACCORDING TO DIAGNOSIS OF PULMONARY EMBOLISM

	Pulmonary Embolism	No Pulmonary Embolism
D-dimer ≥ 500 $\mu\text{g/L}$	195	278
D-dimer < 500 $\mu\text{g/L}$	1	197
Total	196	475
	Sensitivity = 99.5%	Specificity = 41.4%
95% CI	97.2–100.0	37.0–45.9

tent with that in other series reported in the literature (4, 6). Table 1 summarizes the results of the diagnostic strategy used in the study.

Sensitivity of D-dimer (Cutoff Level of 500 $\mu\text{g/L}$)

The diagnostic sensitivity of the D-dimer concentration was 99.5% (95% CI: 97.2 to 100.0), the plasma level being above the cutoff value (500 $\mu\text{g/L}$) in 195 of 196 patients with PE (Table 2). One patient, who had a negative D-dimer assay result despite a high-probability lung scan and deep-vein thrombosis shown by ultrasonography, was considered as having PE and a false-negative D-dimer assay, and was anticoagulated. Thus, of the 198 patients with a D-dimer concentration below the cutoff value, 197 were free of PE, yielding a negative predictive value of 99.5% (95% CI: 97.2 to 100).

Specificity of D-dimer (Cutoff Level of 500 $\mu\text{g/L}$)

Among the entire study population, the diagnostic specificity of the D-dimer concentration was 41.4% (95% CI: 37.0 to 45.9), meaning that the D-dimer concentration was normal (< 500 $\mu\text{g/L}$) in 196 of the 475 patients without PE, excluding PE in 29% of the entire cohort. If the D-dimer concentration had been measured only in patients with a negative result of lower-limb ultrasonography and a nondiagnostic lung scan, it would have excluded PE in 127 patients (Table 1) (i.e., 19% of the study population). As shown in Figure 1, the specificity of the D-dimer concentration was greatly influenced by age, with a maximum value of 72% in the 30- to 39-yr-old group, and a value as low as 9% in the group above 80 yr of age. Consequently, D-dimer assay would allow excluding PE in two of three outpatients under 60 yr of age who were without PE, but in only one of five patients older than 60 yr.

Operating Characteristics of D-dimer

Figure 2 shows the receiver operating characteristics (ROC) curve obtained for various cutoff values of D-dimer in this study. This curve confirms that 500 $\mu\text{g/L}$ is the most reasonable cutoff value for excluding PE when using the ELISA technique, yielding a high sensitivity (99.5%) for a 41.4% specificity. On the other hand, the specificity of a D-dimer plasma level at or above 4,000 $\mu\text{g/L}$ is 93.1%, and the sensitivity 49.5%.

Results of Follow-up (Patients without PE)

Follow-up was completed for 469 of the 475 (98.7%) patients considered to be without PE according to the diagnostic strategy. Thirty-two patients (6.7%) died, none of PE, and all had an abnormal D-dimer assay result. Of the 198 patients with a D-dimer level below 500 $\mu\text{g/L}$, 197 were considered free of PE and were not anticoagulated. None of the 191 patients who were successfully followed had a thromboembolic event dur-

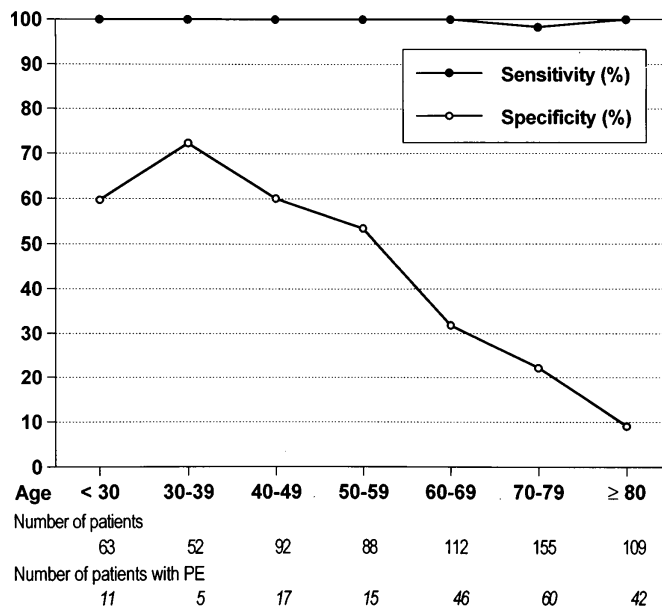


Figure 1. Performance of D-dimer in the diagnosis of pulmonary embolism according to age. Closed circles represents sensitivity, open circles specificity. The numbers below each group represent the number of patients and the proportion of pulmonary embolisms in each age category.

ing the follow-up period. Among the six patients lost to follow-up, four had a normal lung scan and one a normal pulmonary angiogram. Thus, only one patient lost to follow-up (with a low clinical probability of PE and a low-probability lung scan) might have had a false-negative D-dimer test. This would lower the sensitivity only slightly, to 99.0% (95% CI: 96.4 to 99.9), and the negative predictive value to 99% (196 of 198 cases [95% CI: 96.4 to 99.9]), whereas the specificity would be

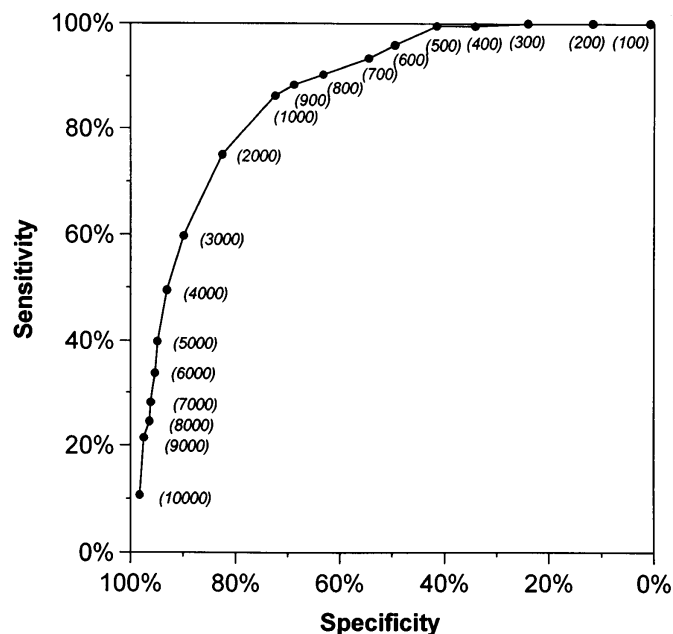


Figure 2. Receiver Operating Characteristics (ROC) curve for plasma D-dimer in the diagnosis of pulmonary embolism. Figures in parentheses represent the respective plasma D-dimer level cutoff values.

essentially unaffected. Hence, withholding anticoagulation from patients with a D-dimer level below 500 $\mu\text{g/L}$ is associated with a conservative 1% risk of thromboembolic events during follow-up (95% CI: 0.1 to 3.6).

DISCUSSION

This was the first large scale study in which the results of an ELISA for D-dimer were used in clinical decision-making for consecutive, unselected outpatients with suspected PE, anticoagulation being withheld from patients with a plasma D-dimer concentration below the cutoff value of 500 $\mu\text{g/L}$ (normal D-dimer). D-dimer assay allowed the exclusion of PE in 29% of patients suspected of having the disease, with an extremely low risk of venous thromboembolism during a 3-mo follow-up.

The present study also provides additional information about two practical characteristics of the D-dimer assay. First, it is definitely more useful in younger patients, with a mean specificity of approximately 60%, in patients under 60 yr of age (Figure 1). In elderly patients, comorbid conditions are likely to increase D-dimer levels with advancing age (22, 23).

Second, the potential use of high plasma levels of D-dimer for ruling in PE as a diagnosis may have been underestimated. Previous studies (14, 15) have insisted on the low capability of D-dimer for establishing the diagnosis of PE, and this is indeed correct at the usual cutoff value of 500 $\mu\text{g/L}$. However, at a cutoff value above 4,000 $\mu\text{g/L}$, the false-positive rate of the D-dimer assay was only 7% in this outpatient series (Figure 2), and 50% of patients with PE exhibited such a value. According to Bayes' rule (20), the probability of PE with a D-dimer level above 4,000 $\mu\text{g/L}$ would be 88% in a patient with a 50% clinical probability of PE and 97% in a patient with an 80% clinical probability of PE. On the other hand, such a result would not be sufficient evidence of PE in a patient with a low clinical probability of having the disease. For example, in a patient with a clinical probability of 20%, the posttest probability of PE would be only 64%. These findings should be confirmed by other investigators before this property of the D-dimer test is used in outpatients suspected of having PE.

Utmost care was taken to avoid common biases in a diagnostic test validation study (15). The patients in the present trial were an unselected outpatient population, presenting the entire spectrum of manifestations of PE: the proportion of high-probability, nondiagnostic, and normal lung scans in the trial was identical to that in the PIOPED study (4). In addition, most pathologic pulmonary angiograms showed only small emboli (segmental or subsegmental). Hence, the sensitivity of the D-dimer assay for PE is unlikely to have been overestimated as a result of spectrum bias (15). Moreover, D-dimer was measured daily by the hemostasis laboratory of our institution in a blinded fashion.

Our conclusions might yet be challenged on the basis of two potential limitations. First, pulmonary angiography was not performed on all of our patients, with the result that the diagnostic standard for classifying patients as having or not having PE may be questioned. However, pulmonary angiography is no longer considered the single diagnostic criterion for PE, and has not been performed systematically in recent studies (6, 15, 19). Lung-scan findings, either normal or of high probability for PE, and deep-vein thrombosis in the presence of an abnormal although nondiagnostic lung scan, are acceptable diagnostic criteria for the disease. Since these criteria were not met by all patients in this series, follow-up was crucial, particularly for patients in whom PE was ruled out by a normal D-dimer assay result. Indeed, in such an outcome study, the negative predictive value of the D-dimer assay re-

sult cannot be calculated from the discharge diagnosis alone, since a normal D-dimer concentration is included in the diagnostic criteria. Negative predictive value must be computed from the follow-up data, with patients presenting a thromboembolic event during the 3-mo follow-up period being considered as having had false-negative results in the D-dimer assay. Moreover, follow-up allows comparison of the risk of thromboembolic events in patients who were not anticoagulated on the sole basis of a normal D-dimer assay result with that in similar outcome studies (6). In the present study, none of the patients who were not anticoagulated on the basis of a normal D-dimer assay result had a thromboembolic event during the follow-up period. Of the six patients without PE who were lost to follow-up, only one failed to meet an accepted diagnostic standard for the absence of PE. Consequently, taking into account both the observed false-negative D-dimer assay result rate in the series, and the potential false-negative result rate from inclusion of this patient, the highest 3-mo risk of PE for patients with a negative D-dimer assay result would be 1.0% (95% CI: 0.1 to 3.6), which is comparable to that for patients discharged on the basis of a normal lung scan (6).

Second, the lack of standardization of the D-dimer assay has hampered its widespread use because of the divergent results obtained with the various available tests. However, the particular ELISA assay used in the present study has consistently shown a very high sensitivity for the presence of acute venous thromboembolism in a number of studies. In a pooled analysis of seven series (13, 23–28) totaling 851 patients suspected of having PE (average prevalence of the disease: 29%) and of the present series, the sensitivity of this ELISA assay for D-dimer was 97% (426 of 440 cases) at a cutoff value of 500 $\mu\text{g/L}$ (95% CI: 95 to 98), setting a standard to which any other commercial test should be compared. Admittedly, this classical ELISA assay is labor intensive and comes in multi-unit batch kits, making it unsuited for emergency use, but a rapid, single, and automated ELISA has recently been developed (29). The diagnostic performance of the new test is identical to that of classical ELISA assays. Novel whole-blood bedside latex assays for D-dimer seem equally promising (30, 31), but the experience with these tests is still limited, and the problem of interobserver variability inherent to the latex technique was not addressed in the studies in which it showed promise. Moreover, the latex assays do not provide quantitative results. Therefore, pending further studies, the conclusions of the present trial cannot be extrapolated to other assays.

In summary, the ELISA for D-dimer used in this study appears to be an extremely useful addition to lower-limb ultrasonography, lung scan, and clinical probability assessment in the workup of outpatients suspected of having PE: at D-dimer concentrations below a cutoff of 500 $\mu\text{g/L}$, the assay allows safe exclusion of the disease in 30% of outpatients suspected of having it, with a very low risk of thromboembolic events during a 3-mo follow-up. This conclusion is now based on a large series of outpatients from whom anticoagulant treatment was withheld when the D-dimer concentration was below the cutoff. The diagnostic efficacy of the assay is greatest in younger patients, and decreases steadily with increasing age. Moreover, D-dimer concentrations above a critical level of 4,000 $\mu\text{g/L}$ may be indicative of PE in patients with a 50% or greater clinical probability of having the disease.

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